Fractional Edge Volume Boxes

Unit 10: Time Capsule Engineers · Lesson 10-2 · EduWonderLab

UNIT 10 · LESSON 10-2 · TEACHER NOTES — **Standard 6.G.2**

FORMAT **Solve task cards**	TIME **25–35 min**
GROUPING **Pairs**	TOPIC **Volume of Rectangular Prisms**
STANDARD **6.G.2**	PREP LEVEL **Light — print prism cards and base-area table**

Learning goal *I can find the volume of a prism with fractional edge lengths.*

Teacher setup

- Print one card set per pair. Each card has at least one fractional edge.
- Students find the base area, then multiply by the height.
- Show why fractional dimensions can still form a real volume.

Materials

Prism cards with fractional dimensions, Base-area table, Fraction multiplication support box, Pencil.

Differentiation & language support

- Support box for multiplying fractions (SPED).
- Sentence stem: “The base area is ___, so the volume is ___.”
- ESOL: relate 1/2 unit to half a cube.

Key vocabulary (English / Español):

Term	Español	What it means
base area	área de la base	length × width of the bottom
edge length	longitud de arista	the length of one side

Quick teacher check: *Check the fraction multiplication on Card 6 (1 1/2 × 1 1/2 × 2).*

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Name: **Date:** **Class:**

Your goal: *I can find the volume of a prism with fractional edge lengths.*

What you need

Prism cards, Base-area table.

How to play

1. Pick a prism card.
2. Find the base area by multiplying length × width.
3. Multiply by the height for the volume.
4. Record the volume in cubic units.

Prism Cards

Find the base area (length × width), then multiply by the height. Multiply fractions carefully.

1. 1/2 × 1/2 × 1/2	**2.** 3/4 × 2 × 2
3. 1 1/2 × 2 × 4	**4.** 1/2 × 3 × 4
5. 2/3 × 3 × 5	**6.** 1 1/2 × 1 1/2 × 2
7. 3/4 × 4 × 2	**8.** 2 1/2 × 2 × 2

Record your work (base area × height):

Explain your thinking

Explain how a prism with fractional edge lengths can still have a whole-number volume.

Sentence frame: *The base area is ___, so the volume is ___ × ___ = ___.*

★ **Challenge:** *Card 1 gives 1/8 cubic unit. How many of those small prisms fill one whole cubic unit? (8)*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. 1/2 × 1/2 × 1/2 **Answer: 1/8 cubic unit**	**2.** 3/4 × 2 × 2 **Answer: 3 cubic units**
3. 1 1/2 × 2 × 4 **Answer: 12 cubic units**	**4.** 1/2 × 3 × 4 **Answer: 6 cubic units**
5. 2/3 × 3 × 5 **Answer: 10 cubic units**	**6.** 1 1/2 × 1 1/2 × 2 **Answer: 4 1/2 cubic units**
7. 3/4 × 4 × 2 **Answer: 6 cubic units**	**8.** 2 1/2 × 2 × 2 **Answer: 10 cubic units**

Teaching notes & look-fors

- Volume = base area × height; convert mixed numbers to improper fractions first.
- Card 6: 3/2 × 3/2 × 2 = 9/2 = 4 1/2.